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Project Report

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IoT-Enabled Pet Feeder Dispenser with Environmental Control Using ESP32 and Blynk IoT Platform

Abstract

This project introduces an Internet of Things (IoT)-based Pet Feeder Dispenser with Environmental Control, designed to address the challenges faced by pet owners in ensuring timely feeding and monitoring of their pets' environment. Using an ESP32 microcontroller as the central processing unit, the system automates pet feeding based on pre-defined schedules while allowing remote control via the Blynk IoT platform. The integration of ultrasonic sensors enables accurate monitoring of the remaining food quantity, while PIR motion sensors detect pet activity to trigger additional feeding if needed. Environmental control is achieved through a DC motor (for ventilation) and a buzzer for audio alerts. The 16x2 LCD provides on-site feedback, while real-time updates are sent to the user's smartphone. This dual control (automatic + manual) approach ensures flexibility, convenience, and reliability in pet care. The system's modular design allows future upgrades, such as camera integration and AI-based feeding prediction.

Introduction

Pet owners are increasingly seeking automation solutions to manage daily responsibilities, especially when they cannot physically attend to their animals. Feeding pets on time is crucial for maintaining their health, yet busy schedules, work commitments, or travel can disrupt consistent feeding patterns. Traditional pet feeders, while helpful, often lack real-time monitoring, environmental sensing, or remote accessibility.

This project addresses these shortcomings by developing an IoT-based solution that combines scheduled automation, sensor-based feeding triggers, environmental control, and smartphone-based monitoring. The ESP32 microcontroller, chosen for its built-in WiFi capabilities and low power consumption, acts as the central hub. It connects with multiple sensors and actuators to create a fully functional smart pet feeder.

By integrating with the Blynk IoT platform, the system enables users to:

i. Schedule or trigger feeding remotely.
ii. Monitor food levels in real time.
iii. Receive alerts when motion is detected.
iv. Control environmental conditions around the feeding area.

The end result is a smart, adaptive, and user-friendly system that minimizes human effort while ensuring the pet's needs are met.

Background Study

Pet feeding automation has evolved from simple mechanical dispensers into intelligent, IoT-enabled systems capable of remote monitoring and control. Traditional feeders, which operated using fixed timers, offered basic convenience but lacked adaptability to pets' real-time needs and provided no feedback to the owner [2].

The integration of microcontrollers such as Arduino, NodeMCU, and ESP32 has enabled more advanced features, including sensor-based monitoring, scheduled feeding, and remote operation [1][3][4]. With built-in Wi-Fi connectivity, boards like the ESP32 allow direct communication with IoT platforms, enabling real-time data transfer between devices and mobile applications [4]. This connectivity forms the backbone of modern smart pet feeders, allowing owners to control feeding, check status, and receive alerts from virtually anywhere.

Various sensing technologies enhance automation. Ultrasonic sensors are frequently used for non-contact measurement of remaining food quantity [1][4], while Passive Infrared (PIR) sensors detect pet movement or presence in the feeding area [4]. Actuators such as servo motors enable precise control over food dispensing mechanisms [1][4], and DC motors can be applied for functions like ventilation or environmental management [5]. Additional components, such as buzzers and LEDs, provide audible and visual alerts [5].

Several IoT-based designs in recent studies highlight the potential and limitations of current solutions. Jayarathne et al. [1] developed a pet feeder robot with scheduled automation and IoT connectivity, demonstrating the practicality of integrating multiple sensors and actuators. Kondapalli et al. [2] implemented an automatic feeder controlled via the Blynk platform, offering remote control but without environmental monitoring features. Jefferi [3] presented a Blynk-based feeder enabling remote manual control, although it lacked redundant feeding triggers. Purwanto et al. [4] designed an animal feeder using NodeMCU ESP8266 and the Blynk app, showing cost-effectiveness and simplicity but without advanced environmental controls. Abdulla et al. [5] proposed an IoT-based feeder incorporating basic alerting and feeding mechanisms but without local display feedback.

This project builds upon these works by combining scheduled automation, motion-triggered feeding, food-level monitoring via ultrasonic sensing, ventilation control with a DC motor, and dual feedback systems through both an LCD display and smartphone notifications. By leveraging the ESP32's Wi-Fi capabilities and the Blynk IoT platform, the system addresses limitations identified in earlier designs, offering a more comprehensive and adaptable solution for modern pet care.

Proposed Methodology

The methodology involves the integration of hardware components, software logic, and IoT connectivity to create an autonomous yet remotely accessible pet feeder. The approach is divided into the following phases:

1. Hardware Integration

i. **Servo Motors**: One servo rotates the food dispensing mechanism to release food; the other controls ventilation if needed.
ii. **Ultrasonic Sensor**: Measures the distance to the top of the stored food to estimate quantity.
iii. **PIR Sensors**: Two motion detectors provide wide-angle coverage to detect pet presence.
iv. **Buzzer and LED**: Indicate alerts such as motion detection or feeding events.
v. **DC Motor**: Provides ventilation control to maintain air circulation.
vi. **LCD Display**: Shows system status locally for quick checks.

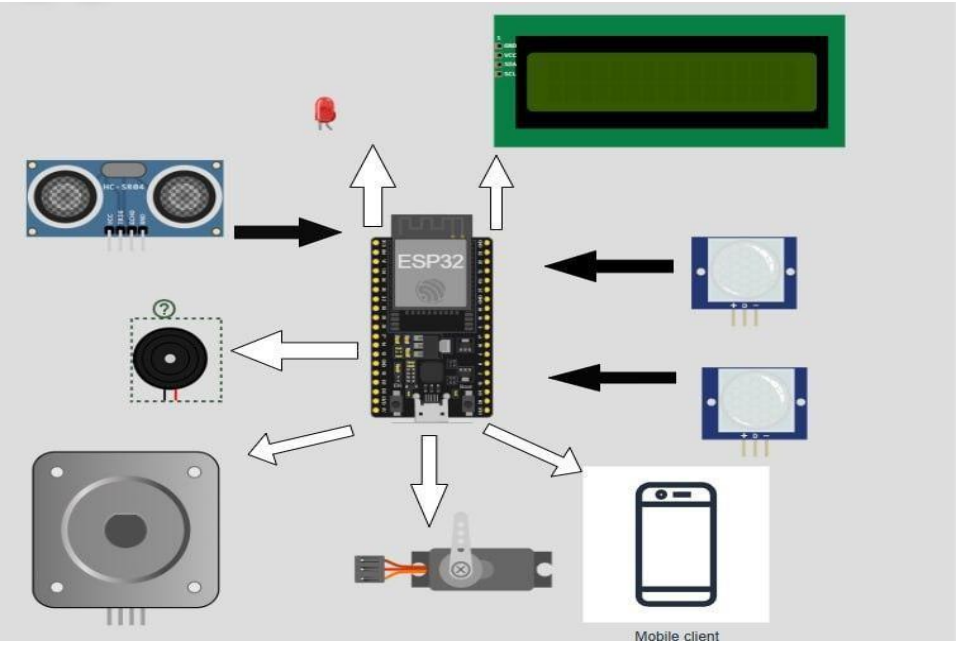

Fig 1: Proposed Block Diagram for Controlling Each Connection

2. Software Development

i. **Arduino IDE**: Used for firmware programming in C++.
ii. **Blynk IoT Platform**: Configured with virtual pins to control functions and display sensor data.
iii. **Timer Functions**: Manage periodic tasks such as checking feeding schedules, monitoring food level, and refreshing display status.

iv. **Time Library**: Synchronizes feeding schedule with real-world time.

3. IoT Connectivity

i. **WiFi Integration**: The ESP32 connects to a local WiFi network for internet-based control.
ii. **Blynk Server Communication**: Data from sensors is sent to the Blynk dashboard, and control commands from the app are executed on the device.

This combination ensures both automation (scheduled logic) and manual override (via smartphone) are possible at any time.

Analysis of System Design

i. **Input Layer Sensors:** An ultrasonic sensor measures the remaining food by checking the distance inside the container, while PIR sensors detect any movement in the feeding area.

ii. **Processing Layer ESP32:** The ESP32 processes the sensor data, checks it against set limits (for example, if motion lasts more than 5 minutes), and matches the current time with the feeding schedule. It also responds to manual control commands sent through the Blynk app.

iii. **Output Layer Actuators & Display:** A servo motor dispenses food, a buzzer gives sound alerts, and an LED provides visual signals. A DC motor runs the ventilation system, and an LCD screen displays system status and sensor readings.

iv. **IoT Layer Remote Access:** Through the Blynk mobile app, the system can be monitored and controlled in real time, with alerts sent directly to the user.

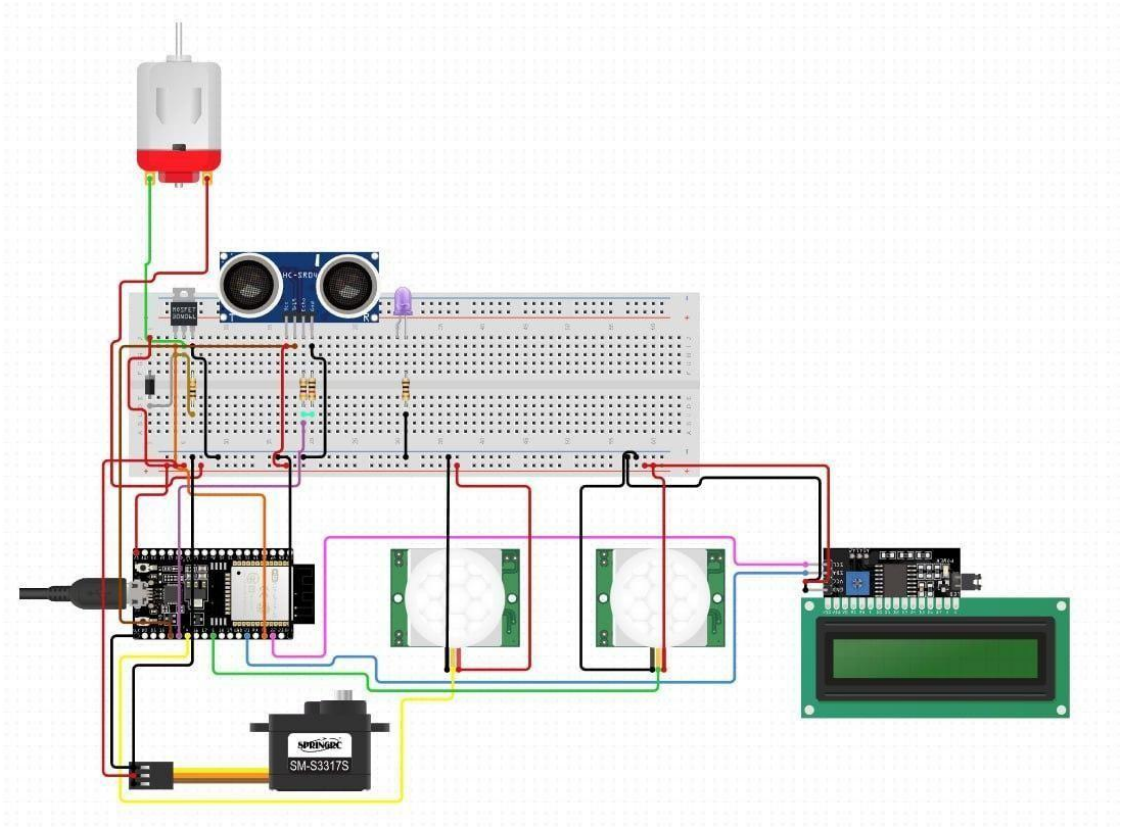

Fig 2: Circuit Diagram for Controlling Each Connection.

The system incorporates redundancy by supporting both scheduled automation and manual control, ensuring uninterrupted operation under various conditions. If the predefined feeding schedule fails due to timing conflicts, sensor errors, or network delays, the manual override accessible through the Blynk mobile app—can be used to dispense food instantly. This dual-mode functionality not only safeguards against missed feedings but also provides flexibility for pet owners to adjust feeding times or quantities in response to unforeseen circumstances. Furthermore, the inclusion of multiple triggering mechanisms (time-based and motion-based) enhances reliability, ensuring that pets are fed even in the event of partial system malfunction or sensor anomalies.

Collecting Hardware Requirements

The hardware components for the system were carefully chosen to achieve an optimal balance between cost-effectiveness, functionality, and ease of integration. Each element was selected based on its proven reliability in IoT-based automation projects, compatibility with the ESP32 microcontroller, and ability to operate efficiently in continuous or intermittent duty cycles.

Components	Purpose
ESP32	Central control with WiFi capabilities
Servo Motor	Dispensing food
PIR Sensors (x2)	Detecting pet presence/activity
DC Motor	Environmental ventilation
Buzzer	Audible alerts
LED Indicator	Visual motion/alert indicator
16x2 I2C LCD Display	Local feedback display
Power Supply (5V)	Powering all components
Breadboard & Wires	Circuit prototyping
Ultrasonic Sensor	Measuring remaining food quantity

Arrangement of Software Requirement

The software environment consists of:

i. **Arduino IDE**: Used for code development, uploading firmware, and serial debugging.
ii. **ESP32 Board Package**: Installed in Arduino IDE to compile and upload code to ESP32.
iii. **Blynk IoT Platform**: Configured with a unique template ID, device name, and authentication token.
iv. **Required Libraries**:
    - WiFi.h – WiFi connection handling.
    - BlynkSimpleEsp32.h – Communication with Blynk server.
    - ESP32Servo.h – Controlling servo motors.
    - NewPing.h – Ultrasonic sensor control.
    - LiquidCrystal_I2C.h – LCD handling via I2C.
    - TimeLib.h – Time management for scheduling.
    - Wire.h – I2C communication.

Working Procedure of System Model

i. At startup, the ESP32 initializes all connected peripherals and establishes both the Wi-Fi connection and the link to the Blynk IoT platform. The LCD displays initialization messages to confirm system readiness.

ii. For scheduled feeding, the system checks the current time against predefined feeding slots every second. If a match is found and the feeding for that slot has not yet occurred, the servo motor activates to dispense food.

iii. Manual feeding is supported through the Blynk application. A dedicated button on virtual pin V1 allows the user to trigger the servo at any time.

iv. The food level is monitored every three seconds using the ultrasonic sensor. Measurements are shown on the LCD and simultaneously updated in the Blynk dashboard.

v. Motion-based feeding uses two PIR sensors to detect pet activity. If continuous movement is detected for more than five minutes, the system automatically dispenses food.

vi. Environmental control is managed via the Blynk interface, enabling remote activation of the buzzer for alerts and operation of the DC motor for ventilation.

vii. The LCD continuously displays the feeding status, food-level readings, and the operational state of connected devices, ensuring clear and real-time system feedback.

Comparison with IoT and Non-IoT System

Feature	**IoT-Based**	**Non-IoT**
Remote Control	Yes	No
Real-Time Status	Yes	No
Scheduled Feeding	Yes	Yes
Motion Detection Feeding	Yes	Rare
Alerts via Phone	Yes	No
Environmental Control	Yes	Manual only

Real Time Analysis

i. During live testing, feeding events were executed precisely at the programmed times without any noticeable delay.
ii. The ultrasonic sensor consistently provided food-level measurements with an accuracy of approximately ±1 cm, ensuring reliable monitoring of remaining food quantities.
iii. The PIR motion sensors successfully detected pet movement within a 120° field of view, providing effective coverage of the feeding area.
iv. Data updates from the ESP32 to the Blynk dashboard occurred within one to two seconds of each sensor reading, enabling near real-time remote monitoring.
v. Motion-based feeding was reliably activated after five minutes of continuous pet presence, confirming the effectiveness of the activity-triggered feeding mechanism.
vi. These results demonstrate that the system functions effectively in real-world conditions, assuming a stable internet connection is maintained.
vii. This confirms the system operates effectively in real-world conditions, provided a stable internet connection exists.

Discussion and Limitations

i. The system successfully achieves its intended objectives, delivering automated feeding, real-time monitoring, and environmental control. However, several limitations affect its performance in certain scenarios.

ii. First, the system is dependent on a stable internet connection. In the absence of Wi-Fi, all remote control and monitoring features become unavailable, limiting the system to local operation only.

iii. Second, the design currently lacks a backup power source. Power outages immediately halt all functionality, potentially disrupting scheduled feeding cycles.

iv. Finally, the ultrasonic sensor's performance can be influenced by uneven food surfaces, which may cause minor fluctuations in measurement accuracy.

v. Despite these challenges, the system's modular architecture allows for straightforward upgrades. Future enhancements, such as offline control modes, battery backup integration, and sensor calibration improvements, can address these limitations effectively.

Applications

i. The system is well-suited for domestic pet care, particularly for owners with demanding schedules who require automated feeding and environmental monitoring.

ii. It can be deployed in animal shelters that manage multiple feeding zones, ensuring consistent and timely feeding for all animals.

iii. The design is also adaptable for use in zoos or research facilities, especially for small mammals and birds, where controlled feeding schedules are essential for animal welfare.

iv. In veterinary recovery wards, the system can assist in providing regulated feeding for animals undergoing treatment or rehabilitation, reducing the need for constant manual intervention.

Limitations and Future Improvements

i. The system's IoT functionality is dependent on an active internet connection. Without Wi-Fi access, remote monitoring and control features become unavailable, restricting operation to local control only.
ii. The current design dispenses a fixed portion of food during each feeding cycle, which may not be suitable for pets with varying dietary needs.
iii. Additionally, the system lacks live video monitoring capabilities, limiting the owner's ability to visually confirm feeding events or observe pet behavior in real time.

Future Improvements:

i. Future enhancements to the system could include the integration of a load cell to measure food weight, enabling precise portion control during each feeding cycle.
ii. A camera module could be incorporated to provide live visual monitoring, allowing pet owners to observe feeding events and their pet's behavior remotely.
iii. Adding a battery backup would ensure uninterrupted operation during power outages, maintaining both scheduled and manual feeding capabilities.
iv. The implementation of AI algorithms could enable adaptive feeding schedules based on activity trends detected by the motion sensors, providing a more personalized feeding approach for individual pets.

Conclusion

The IoT Pet Feeder Dispenser with Environmental Control successfully automates pet feeding while allowing remote supervision and manual intervention. It combines automation, real-time monitoring, and environmental management in one system. The project demonstrates how IoT can be applied to everyday challenges, improving convenience and ensuring pet welfare. With further enhancements like AI and video monitoring, it has the potential to become a comprehensive pet care solution.

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